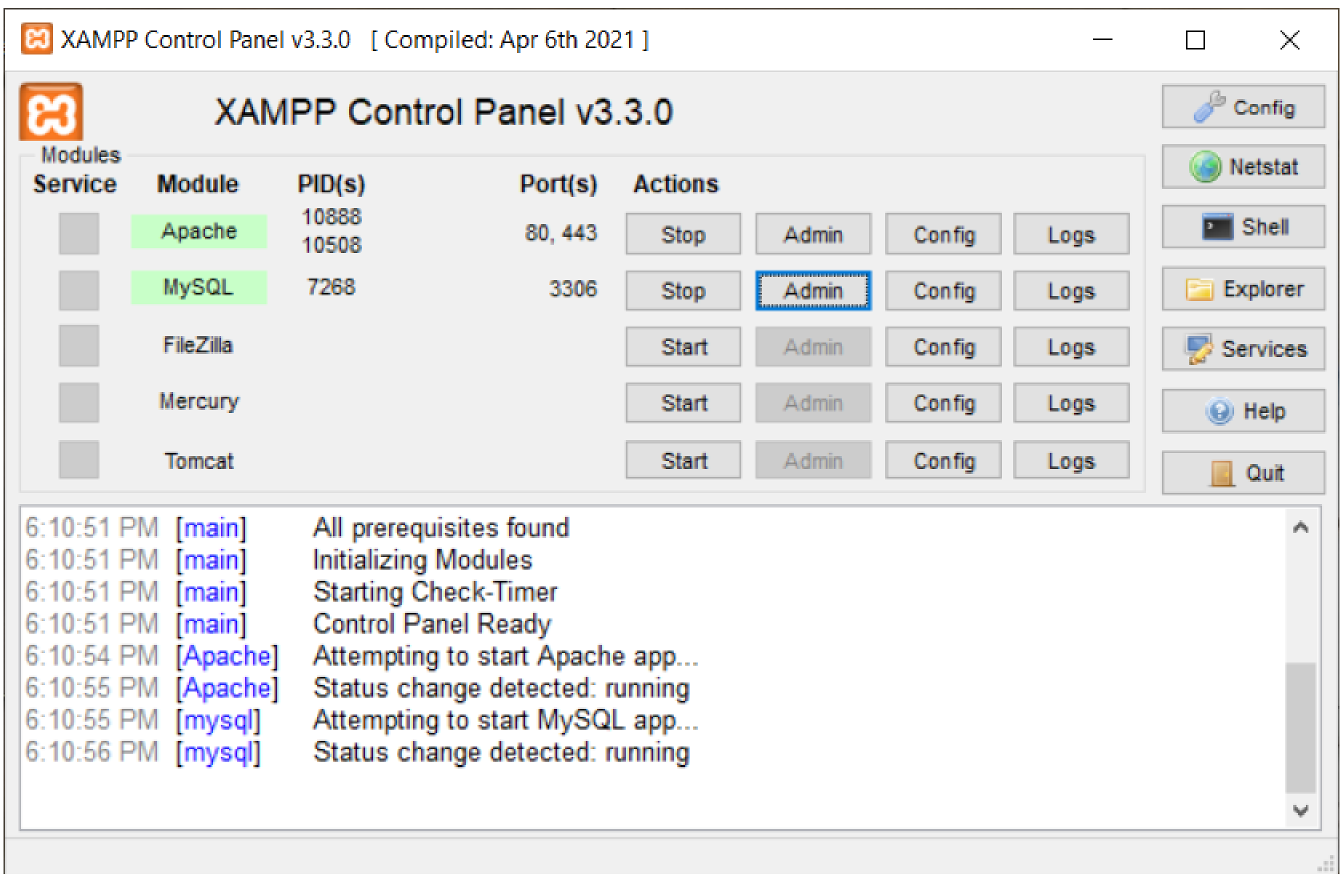
TOPSTECHNOLOGIES
Training | Outsourcing | Placement | Study Abroad

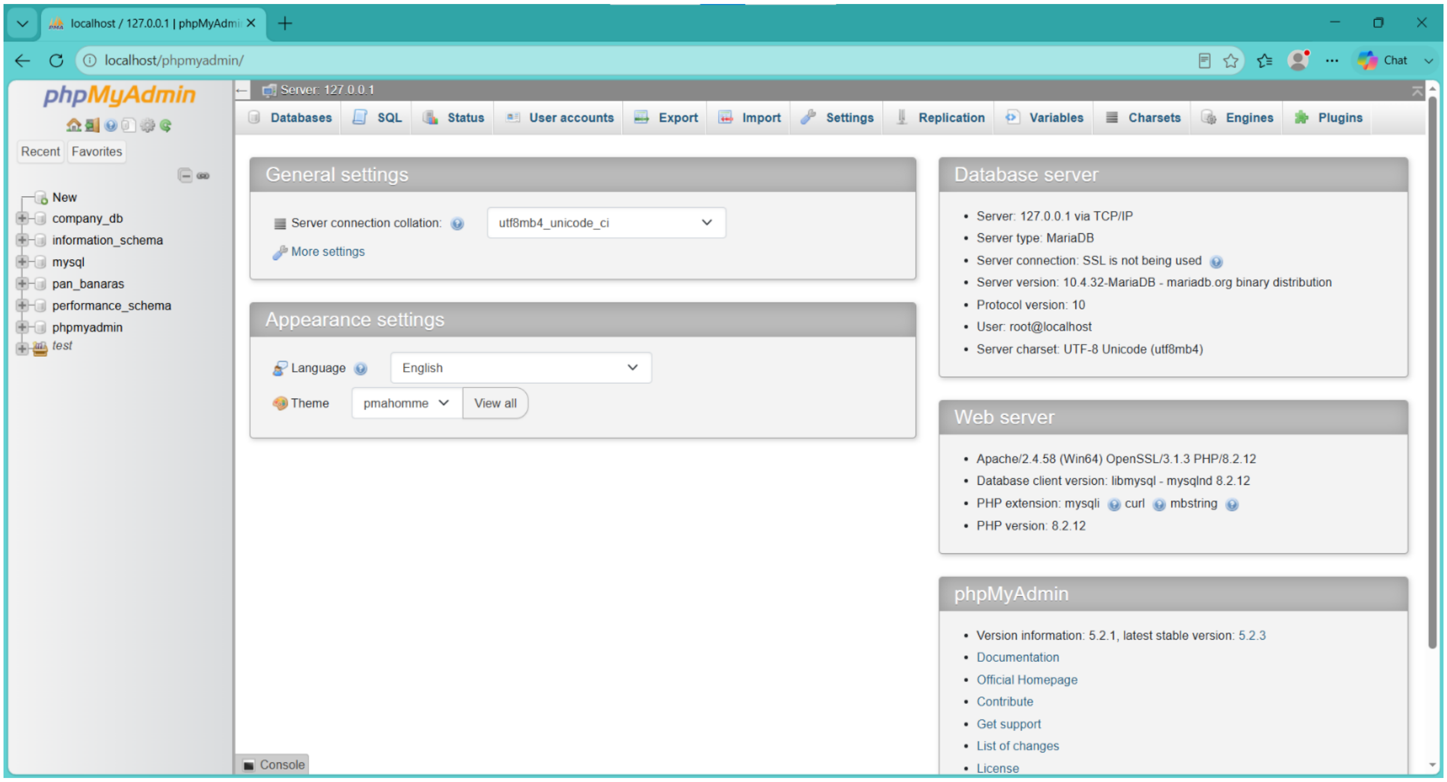
PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 4
Database Management – SQL & PL/SQL
SESSION / ASSIGNMENT – 1

TASKS :

1. **Install MySQL Workbench on your computer, connect to the local MySQL server, and take a screenshot of the default databases shown in the Schemas panel.**



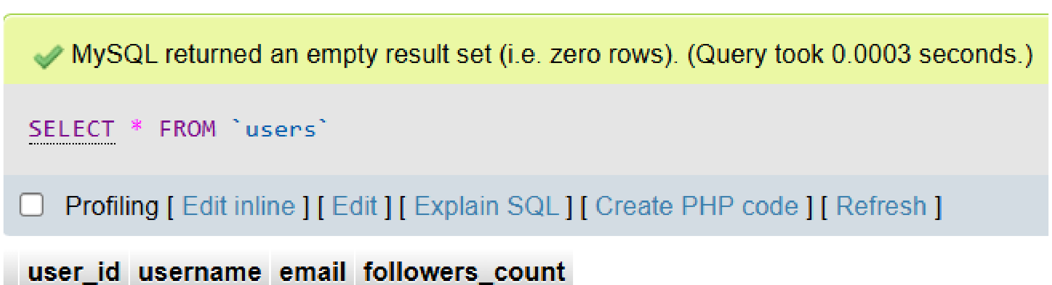


2. **Create a new database called InstaClone in MySQL Workbench, then create a table named Users with columns: user_id (INT, primary key), username (VARCHAR), email (VARCHAR), and followers_count (INT).**

➔ **CODE:**

```
CREATE TABLE Users(
user_id INT PRIMARY KEY,
username VARCHAR(100),
email VARCHAR(100),
followers_count INT );
```

OUTPUT:



3. **Insert 3 sample users into the Users table you created, using realistic Instagram-style usernames and follower counts.**

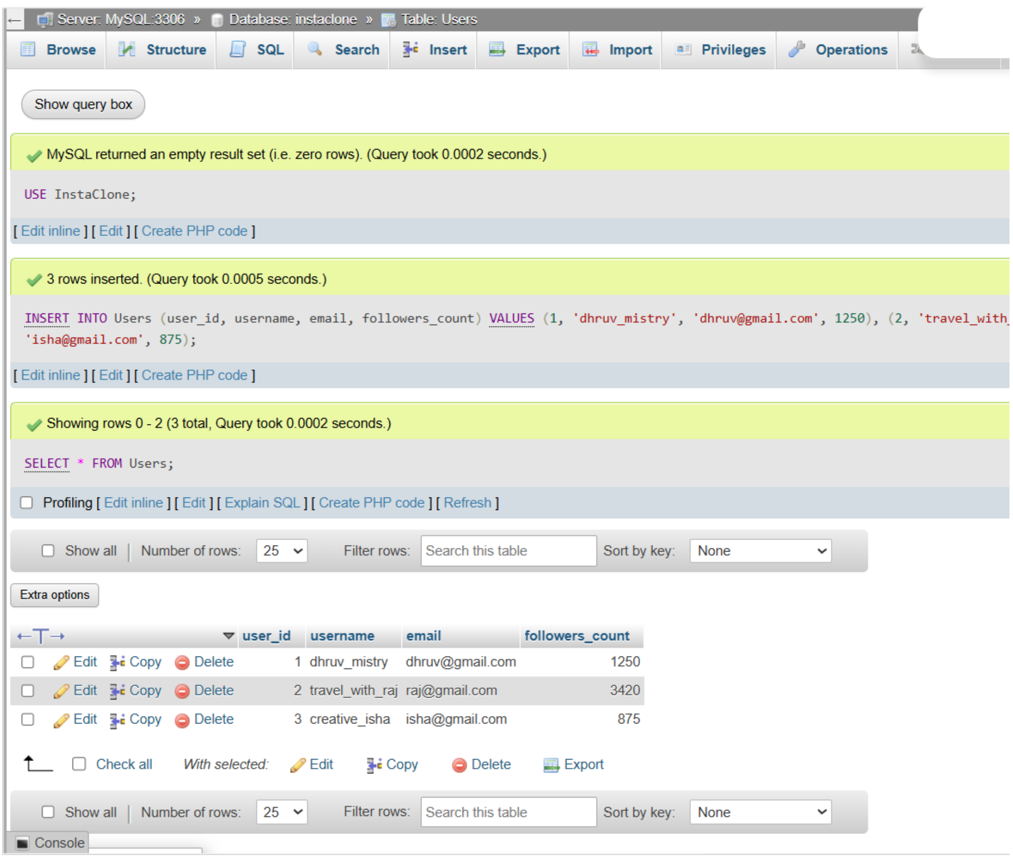
➔ **CODE:**

```
Use InstaClone;

INSERT INTO Users (user_id, username, email, followers_count)
VALUES
(1, 'dhruv_mistry', 'dhruv@gmail.com', 1250),
(2, 'travel_with_raj', 'raj@gmail.com', 3420),
(3, 'creative_isha', 'isha@gmail.com', 875);

SELECT * FROM Users;
```

OUTPUT:



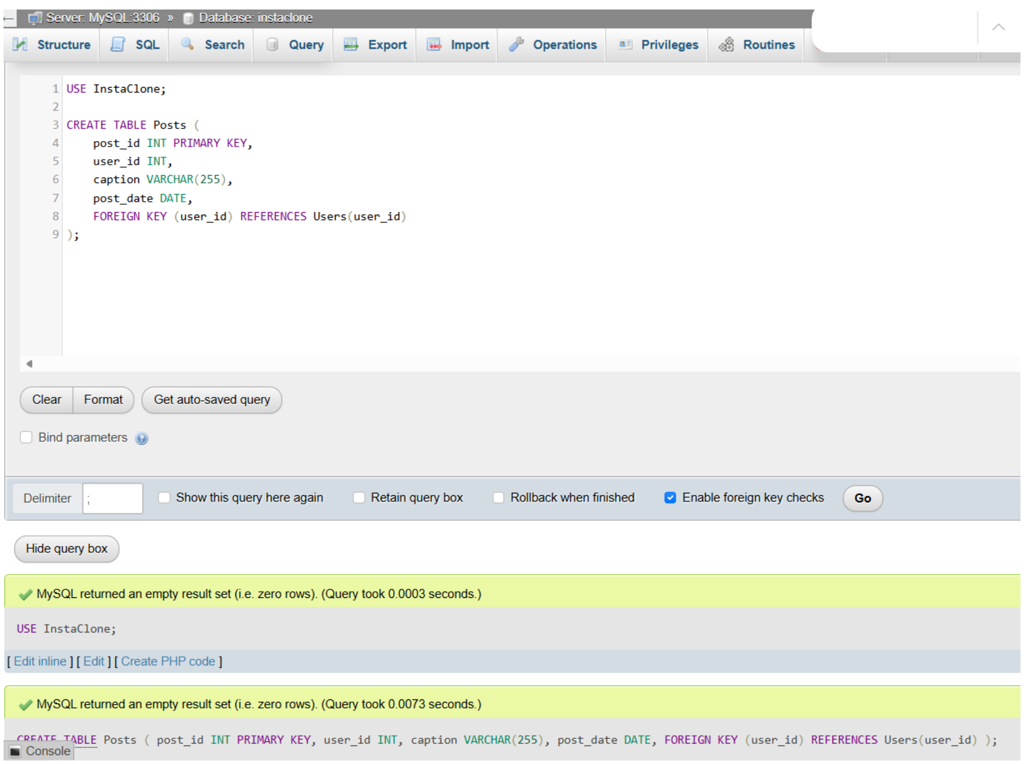
4. **Create another table in the InstaClone database called Posts with columns: post_id (INT, primary key), user_id (INT), caption (VARCHAR), and post_date (DATE). Add a foreign key from Posts.user_id to Users.user_id to establish a relationship.**

➔ **CODE:**

```
USE InstaClone;

CREATE TABLE Posts (
   post_id INT PRIMARY KEY,
   user_id INT,
   caption VARCHAR(255),
   post_date DATE,
   FOREIGN KEY (user_id) REFERENCES Users(user_id)
);
```

OUTPUT:



5. **Write a short comparison (3-4 lines each) of MySQL, PostgreSQL, Oracle, and SQLite, focusing on where each one is commonly used (for example: web apps, mobile apps, enterprise systems, etc.).**

➔ **1. MySQL**

MySQL is widely used for **web applications and websites**.
It is popular with PHP, WordPress, and other web development technologies.
It is suitable for small to large applications and is easy to use.

2. PostgreSQL

PostgreSQL is an advanced **open-source relational database**.
It is commonly used for web applications, data analysis, and applications requiring complex queries.
It is known for reliability, flexibility, and strong SQL support.

3. Oracle

Oracle Database is mainly used in **large enterprise systems**.
Banks, governments, and large organizations use it for handling huge amounts of critical data.
It provides strong security, scalability, and transaction management.

4. SQLite

SQLite is a lightweight database that does not require a separate server.
It is commonly used in **mobile apps, desktop applications, and embedded systems**.
It is especially useful when a simple, local database is required.